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Deployment Optimization for Search Task in Netted Radar System under Region Constraints

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***Abstract*—This paper studies a deployment optimization method of netted radar systems under radar location constraints. The key idea is to improve the search capability by radar deployment optimization. We first adopt the surveillance region as the metric function to gauge targets' searching performance. Then, various constraints are considered and the optimization model is introduced as a nonconvex optimization problem. To solve this problem, we use the simulated annealing (SA) algorithm to acquire the deployment optimization scheme. Finally, simulation results demonstrate that the deployment proposed can enlarge netted radar systems' detection area, which verifies the validity of the radar deployment scheme proposed.**

Keywords—deployment optimization, netted radar system, region constraint, radar search, simulated annealing algorithm.

I. Introduction

Due to the increasingly complex battlefield environment and target characteristics in modern wars, the netted radar system is gaining more and more popularity for its better performance in target search capability and tracking accuracy. Correspondingly, how to improve the radar network performance with limited resources has attracted lots of attention in various application areas [1]-[2].

Some existing research primarily focuses on improving radar detection performance by optimizing transmit parameters such as transmit power, dwell time, and bandwidth [3]-[6]. A resource allocation scheme is built in [3] to balance the resource usage for search and track applications to simultaneously maximize the search capability and the tracking accuracy. Reference [4] proposes a joint target assignment and resource optimization strategy, which achieves better tracking accuracy and low probability of intercept performance by jointly optimizing the target-to-radar assignment, bandwidth, and dwell time allocation. Reference [5] increases the radar network's target capacity by adapting each node's transmit power and dell time. Reference [6] maximizes the tracking performance by optimizing the radar scheduling and waveform correlation matrix of each netted multiple-input multiple-output radar.

Indeed, optimizing the configuration of radars is also an efficient strategy to improve the detection ability of netted radar systems. Some sensor placement optimization work [7–10] provides solutions to the radar network deployment problem. Considering the constraints of sensor geometry, reference [7] determines the optimal sensor placements under position constraints to increase sensor positioning accuracy. The algorithm proposed in [8] focuses on the transmitter's and receivers' placement of distributed MIMO radar systems, which reduces the Cramer-Rao lower bounds of the target and improves the positioning performance. Reference [9] proposes the improved cuckoo search algorithm to solve a multi-sensor deployment detection model to increase the convergence speed and accuracy. However, the above literature mostly considers the location performance when dealing with the placement optimization problem, and the coverage area is less considered. Reference [10] proposes a deployment model for maximum coverage area. But the coverage area considered in [10] is the region intersected by the detecting regions of single sensors, which has errors with the actual coverage area.

To tackle the above problems, a radar deployment optimization method for searching task is given in this paper. We first obtain the actual detecting regions of the netted radar system according to the detection probability (PD). The intersection area between the detecting region and the interest region is used as the metric function. Then, we formulate the deployment optimization model under position constraints as a nonconvex optimization problem and apply the SA algorithm to solve it. Finally, the simulation results verify the superiority of the deployment optimization scheme compared with the random deployment scheme.

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II. SYSTEM MODEL

As shown in Fig. 1, consider a netted radar system with N radar nodes to detect the interest region. Let the interest area be I and the location of the ith radar node is defined as $(x_i, y_i), i = 1, 2, \cdots, N$.

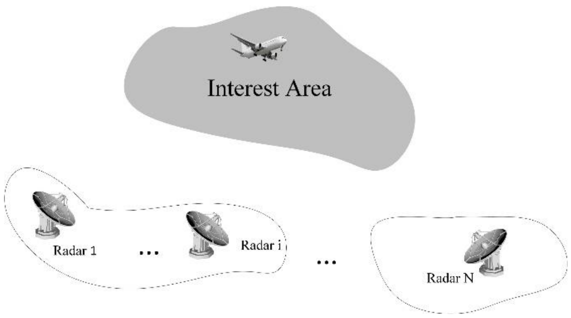


Fig. 1. The deployment under the radar position constrain

We assume that the detection region of this netted radar system is D, which is mostly considered as the coverage area intersected by the detecting regions of single radar stations[10]. In fact, D is the coverage in the sense of the PD according to the different fusion rules [11]. Let F be the effective detection region of the radar system, which is the area overlapped by the interest region and the detection region, as follows:

$$F(\mathbf{x}) = I \cap D(\mathbf{x}) \tag{1}$$

where $\mathbf{x} = [x_1, y_1, \cdots, x_N, y_N]$ is the deployment scheme of each radar.

Let $R_{i\max}$ be the maximum detecting range of the ith radar node, which can be given by the radar equation as follows:

$$(R^4)_{i\max} = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 K T_0 B F_n L_0 SNR_i} \tag{2}$$

where P_t presents radar transmitting power, σ is RCS, G_t and G_r are the radar transmitting antenna and receiving antenna gain, respectively. In (2), λ is radar transmitted wavelength, K is Boltzmann constant with the value of $1.38 \times 10^{-33}\,\mathrm{W \cdot s}$, B denotes receiver bandwidth, F_n is noise factor and L_0 is system loss factor. T_0 denotes the receiver noise temperature and its magnitude is usually taken as 290K. The corresponding p_{d0} of this radar is calculated as (3) [12].

$$p_{d0} = \exp\left(-\frac{4.75}{SNR}\right) \tag{3}$$

We use the Monte Carlo method to estimate $F(\mathbf{x})$. At first, we are supposed to get the PD of a single radar. Take a point P as an example, whose position is (x_q, y_q). The distance between the point P and the ith radar node is as follows:

$$R_i = \sqrt{(x_q - x_i)^2 + (y_q - y_i)^2} \tag{4}$$

Substituting p_{d0}, R_i, and $R_{i\max}$ into (5), the detection probability of the ith radar is given.

$$p_{di} = \exp\left(\left(\frac{R_i}{R_{\max}}\right)^4 \ln p_{d0}\right) \tag{5}$$

Then, the fusion PD of the netted radar system can be written as (6), where $f(\cdot)$ denotes the fusion criterion.

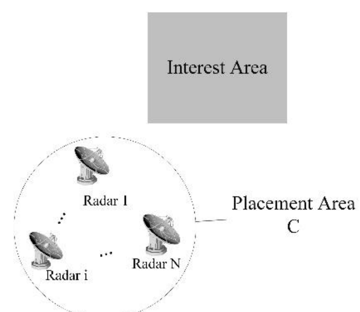
$$Pd_{fusion} = f(p_{di}), i = 1, \cdots, N \tag{6}$$

Once the global PD of P has been determined, this can be compared with the required threshold, $Pd_{threshold}$.

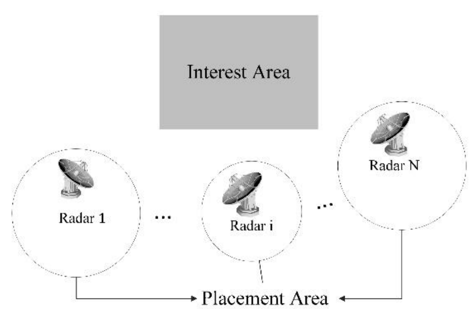
We select M points from I randomly. Assuming that the number of points that satisfy $Pd_{fusion} > Pd_{threshold}$ be N_t. Thus, $F(\mathbf{x})$ is approximated as

$$F(\mathbf{x}) = I \times \frac{N_t}{M} \tag{7}$$

In practical situations, the deployment of radars is bound to be subject to natural features as well as other factors. To this end, there are two constrained placement areas considered in this paper, as shown in Fig. 2. Fig. 2(a) shows the scenario that N radar stations deploy inside the circle with radius ρ_0 whose center is (X_0, Y_0), that is, (x_i, y_i) satisfy the condition $(x_i - X_0)^2 + (y_i - Y_0)^2 \le \rho_0^2, i = 1, \cdots, N$. Fig. 2(b) means that position of ith radar (x_i, y_i) satisfy $(x_i - X_i)^2 + (y_i - Y_i)^2 \le \rho_i^2, i = 1, \cdots, N$.



(a) Radars are located in the same placement area



(b) Radars are located in different placement areas

Fig. 2. The constrained placement area of radars

III. THE DEPLOYMENT OPTIMIZATION OF NETTED RADAR SYSTEMS

Assume that the placement region is C and the *ith* radar is constrained to be deployed in C_i. Thus, the constrain of the placement area proposed in section II can be simplified as:

$$(x_i, y_i) \in C_i \tag{8}$$

It is clear that the purpose of radar deployment optimization is to find the deployment scheme which maximizes $F(\mathbf{x})$. After the introduction of the constrained conditions and the optimization objectives, the radar deployment optimization problem can be formulated as

$$\begin{aligned} &\max \quad F(\mathbf{x}) \\ &s.t. \quad (x_i, y_i) \in C_i \quad i = 1, \cdots, N \end{aligned} \tag{9}$$

where $F(\mathbf{x})$ denotes the maximum coverage of the detection area, and the constrain denotes the region constraints of the netted radar system.

The proposed mathematical model in (9) is nonconvex. The SA algorithm is used to acquire the deployment optimization scheme in this paper, which can avoid being trapped in local minima. The algorithm sets a certain initial temperature T_0, combines the probabilistic jumping property to continuously search for the global optimal solution in the global solution space with the temperature cooling down. The key computation steps are detailed as follows.

Step (1) Let $T = T_0$. T denotes the initial temperature at which annealing is started.

Step (2) Generate an initial solution $\mathbf{x}_0 = [x_1, y_1, \cdots, x_N, y_N]$ randomly, and calculate the corresponding objective function value $F(\mathbf{x}_0)$.

Step (3) Let $T = kT$, where $k \in \{0,1\}$ is the decreasing rate of the temperature.

Step (4) Apply a random perturbation to the current solution $\mathbf{x}_t$ to generate a new solution $\mathbf{x}_{t+1}$ in its neighborhood. Determine whether $\mathbf{x}_t$ satisfies the constraint. If so, go to step (5); otherwise, re-execute step (4).

Step (5) Compute the corresponding objective function value $F(\mathbf{x}_{t+1})$, and calculate $\Delta F = F(\mathbf{x}_{t+1}) - F(\mathbf{x}_t)$.

Step (6) If $\Delta F \le 0$, accept $\mathbf{x}_{t+1}$ as the current solution. Otherwise, judge whether to accept $\mathbf{x}_{t+1}$ according to the probability $e^{-\Delta F / kT}$.

Step (7) Perform steps (4), (5) and (6) L times at temperature T to repeat the perturbation and acceptance process.

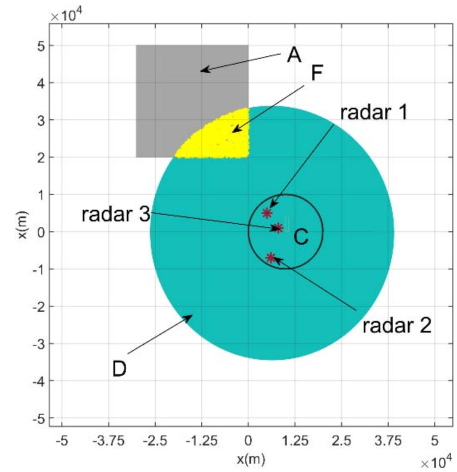
Step (8) Judge whether the temperature reaches the termination temperature level. If so, terminate the algorithm; otherwise, return to step (2).

IV. SIMULATION

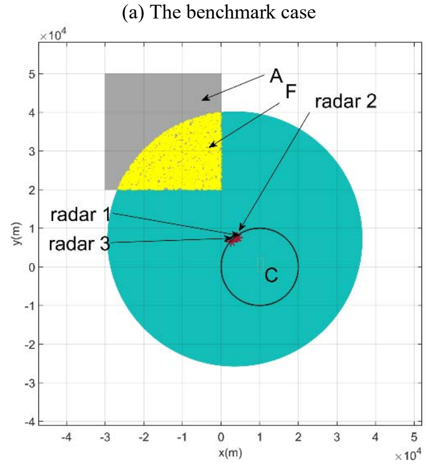
Consider a scenario consisting of $N = 3$ radars. The system parameters of each radar are the same, which are given in Table I. The interest region is set to $x_I \in (\text{-3e4}, 0)\,\mathrm{m}$, $y_I \in (\text{2e4}, \text{5e3})\,\mathrm{m}$, i.e., the gray area shown in Fig. 3 and Fig. 4.

TABLE I. INITIAL PARAMETERS

Parameter	P_t	λ	B	T_0	F_n
Value	33e3 W	0.1 m	1e6 Hz	290 K	6 dB
Parameter	L_0	SNR	RCS	K	G
Value	8 dB	24 dB	1 m^2	1.38e-23 W·s	34 dB



(a) The benchmark case



(b) The optimization case

Fig. 3. The results of Simulation 1

A. *Radars Dep#loy in a Same Placement Area*

For simulation 1, radars all deploy within a circular region, as shown in Fig. 3. The (x_i, y_i) satisfy the condition $(x_i - 1\mathrm{e}4)^2 + y_i^2 \le 1\mathrm{e}4^2$. We set random deployment scheme as a benchmark case, and assume that the positions of radars in the benchmark case are (5e3,5e3) m, (6e3, -7e3) m and (8e3,1e3) m, respectively.

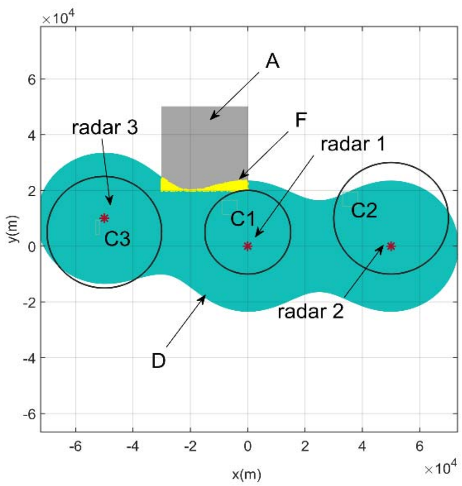
Fig. 3 illustrates the results of simulation 1, where the green region is the search area of this radar system. The F of benchmark case is the area of yellow region shown in Fig. 3(a), with a value of 1.65e7 m^2. The F of the optimization case is 3.87e8 m^2. At this time, the optimized positions are (3.935e3, 7.579e3) m, (4.219e3, 7.587e3) m and (2.458e3, 6.567e3) m.

B. Radars deploy in different placement areas

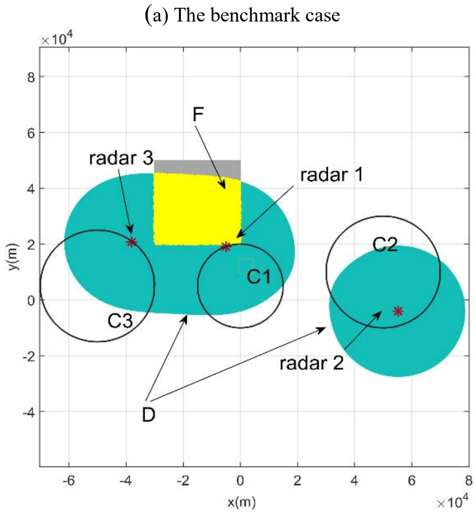
For simulation 2, radars are located in different circular regions. As shown in Fig. 4, the positions $\left(x_i, y_i\right)$ satisfy

$$\begin{cases} \left(x_1-1\mathrm{e}4\right)^2+y_1^2 \leq 1\mathrm{e}4^2 \\ \left(x_2-5\mathrm{e}4\right)^2+\left(y_2-1\mathrm{e}4\right) \leq 2\mathrm{e}4^2 \\ \left(x_3+5\mathrm{e}4\right)^2+\left(y_3-5\mathrm{e}4\right) \leq 2\mathrm{e}4^2 \end{cases} \tag{10}$$

Let the positions of radars in the benchmark case be (0,0) m, (5e4,0) m, and (-5e4,1e4) m.



(a) The benchmark case



(b) The optimization case

Fig. 4. The results of Simulation 2

The result of the benchmark case is given in Fig. 4(a) and F is 5.00e7 m^2. The F of optimization case is 7.44e8 m^2. At this time, the positions are (-0.5047e4, 1.9086e4) m, (5.5218e4, -0.4055e4) m, and (-3.8100e4, 2.0777e4) m. The results verify that deployment optimization scheme can obviously improve the search ability of netted radar systems.

Ideally, the closer the radar is to the target, the better the searching performance of the radar system. Thus, the radars tend to deploy as close to the interest area as possible to maximize the detection region. Considering the constraints of radar placement area, however, radars end to be placed at the boundary of placement area, which is shown in Fig. 4(b). This conclusion is also reflected in Fig. 3(b). Moreover, we find that radar 2 shown in Fig. 4(b) is no longer located on the side of the interest area. The reason is that the placement area of radar 2 is far from the interest area. Radar 2 contributes little to searching the interest area. At this time, the $F(\mathbf{x})$ mainly depends on the positions of radar 1 and radar 3, and radar 2 can be deployed within the placement area randomly.

V. Conclusion

In this paper, a deployment optimization scheme of netted radar systems is built, with the aim of improving search ability by optimizing radars positions. We formulate the deployment optimization under radar position constraints as an optimization model, and use the SA method to solve it. Simulation results demonstrate that the deployment optimization scheme can evidently enlarge the detection region, when compared with the benchmark case. At this time, the radar nodes tend to be located at the boundary of the placement region and close to the interest area.

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